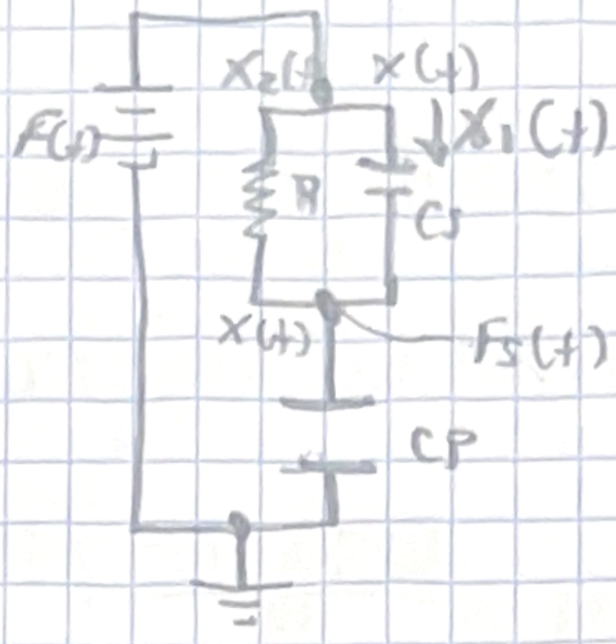
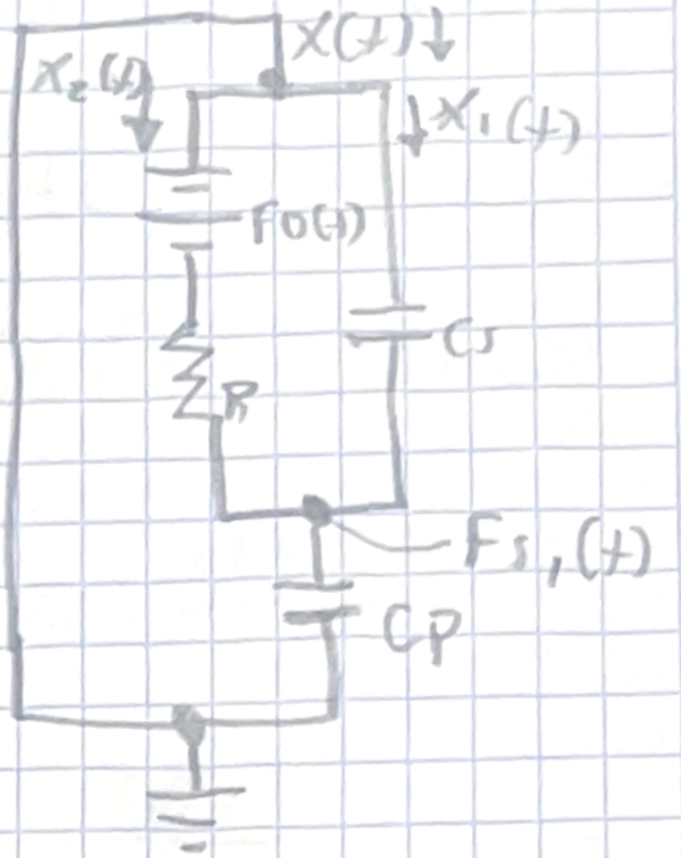
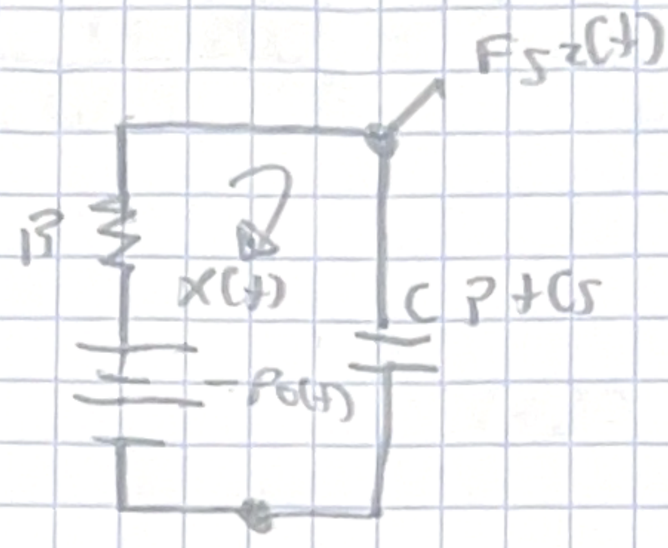
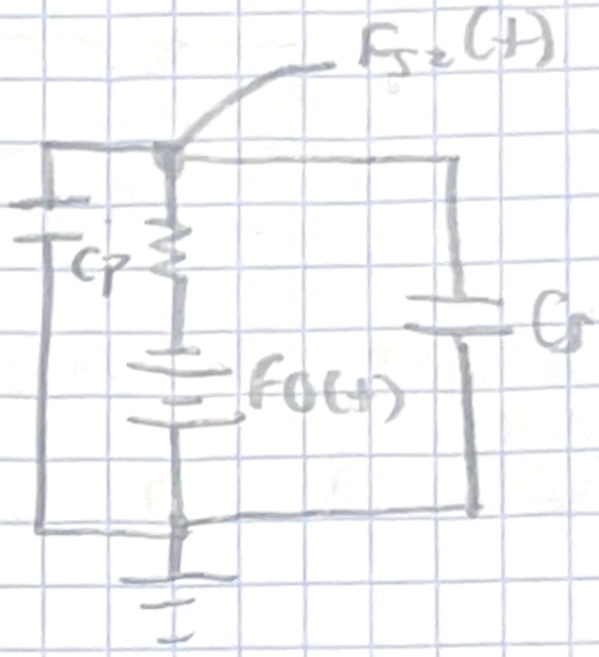
Practica 5.5. "Modelado y control del Sistema musculoesqueletico"











Circuito 1: Apagando $F_0(t)$

$$x(t) = x_1(t) + x_2(t)$$

$$x(t) = C_p \frac{dF_{s1}(t)}{dt}$$

$$x_1(t) = C_s \frac{d[F(t) - F_{s1}(t)}{dt}$$

$$x_2(t) = \frac{F(t) - F_{s1}(t)}{R}$$

$$C_p \frac{dF_{s1}(t)}{dt} = C_s \frac{d[F(t) - F_{s1}(t)}{dt} + \frac{F(t) - F_{s1}(t)}{R}$$

$$C_p s F_{s1}(s) = C_s s [F(s) - C_s s F_{s1}(s) + \frac{F(s)}{R} - \frac{F(s)}{R}$$

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Transformada de Laplace

$$C_2 s F_{s_1}(s) + C_s f_{s_1}(s) + \frac{F_{s_1}(s)}{R} = C_s s F(s) + \frac{F(s)}{R}$$

$$\left(C_2 s + C_s s + \frac{1}{R}\right) F_{s_1}(s) = \left(C_s s + \frac{1}{R}\right) F(s)$$

$$\frac{R(C_2 + C_s)s + 1}{R} F_{s_1}(s) = \frac{C_s R s + 1}{R} F(s)$$

$$F_{s_2}(s) = \frac{\frac{C_s R s + 1}{R}}{\frac{R(C_2 + C_s)s + 1}{R}} = F_{s_1}(s) = \frac{C_s R s + 1}{R(C_2 + C_s)s + 1} F(s)$$

Circuito 2: Apagando $F(t)$

$F_0(t) = \alpha F(t)$

$$-\alpha F(t) = R x(t) + \frac{1}{C_2 + C_s} \int x(t)\, dt$$

$$F_{s_2}(t) = \frac{1}{C_2 + C_s} \int x(t)\, dt$$

$$-\alpha F(s) = R x(s) + \frac{x(s)}{(C_2 + C_s)s}$$

$$F_{s_2}(s) = \frac{x(s)}{(C_2 + C_s)s}$$

$$F(s) = -\frac{1}{\alpha}\left(R + \frac{1}{(C_2 + C_s)s}\right) x(s) = -\frac{1}{\alpha} \frac{R(C_2 + C_s)s + 1}{(C_2 + C_s)s} x(s)$$

$$\frac{F_{s_2}(s)}{F(t)} = \frac{\frac{x(s)}{(C_2 + C_s)s}}{-\frac{1}{\alpha} \frac{R(C_2 + C_s)s + 1}{(C_2 + C_s)s} x(s)} = \frac{\alpha}{R(C_2 + C_s)s + 1}$$

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$$F_{S2}(s) = -\frac{\alpha F(s)}{B(C_p+C_s)s+1}$$

Función de transferencia:

$$F_S(s) = \frac{C_s B s + 1 - \alpha}{B(C_p+C_s)s+1} F(s)$$

$$\boxed{\frac{F_S(s)}{F(s)} = \frac{C_s B s + 1 - \alpha}{B(C_p+C_s)s+1}}$$

* Error en estado estacionario:

$$e(s) = \lim_{s\to 0} F(s)\left(1 - \frac{F_S(s)}{F_S}\right) = \lim_{s\to 0} s\frac{1}{s}\left(1 - \frac{BC_s s + 1 - \alpha}{BC_p + BC_s + 1}\right) = \alpha$$

* Estabilidad en lazo abierto

$$(BC_p + BC_s)s + 1 = 0$$

$$\boxed{\lambda = -\frac{1}{BC_p + BC_s}}$$

El sistema es estable

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